The Frame Problem (McCarthy and Hayes)

- Axioms other than effect axioms are required for formalizing dynamic worlds. These are called *frame axioms*, and they specify the action *invariants* of the domain, i.e., those fluents unaffected by the performance of an action.
  - A positive frame axiom – dropping things does not affect an object's colour:

    $$colour(x, s) = c \supset colour(x, do(drop(r, y), s)) = c.$$

  - A negative frame axiom – not breaking things:

    $$\neg broken(x, s) \wedge [x \neq y \vee \neg fragile(x)] \\ \supset \neg broken(x, do(drop(r, y), s)).$$

- Problem: Vast number of frame axioms; only relatively few actions will affect the truth value of a given fluent. All other actions leave the fluent invariant.
  - An object's colour remains unchanged as a result of picking things up, opening a door, turning on a light, electing a new prime minister of Canada, etc. etc.
- $\sim 2 \times \mathcal{A} \times \mathcal{F}$ frame axioms.
- The *frame problem*:
  - Axiomatizer must think of all these quadratically many frame axioms.
  - System must reason efficiently in the presence of so many axioms.

What Counts as a Solution to the FP?

- Suppose the person responsible for axiomatizing an application domain has suceeded in writing down **all** the effect axioms, i.e. for each relational fluent F and each action A which can cause F's truth value to change, axioms of the form

  $$R(\vec{x}, s) \supset (\neg)F(\vec{x}, do(A, s))$$

  and for each functional fluent f and each action A that can cause f's value to change, axioms of the form:

  $$R(\vec{x}, y, s) \supset f(\vec{x}, do(A, s)) = y.$$

- Want a systematic procedure for generating, from these effect axioms, all the frame axioms.

- If possible, also want a *parsimonious* representation for these frame axioms (because in their simplest form, there are too many of them).

Why Do We Want a Solution to the FP?

- Frame axioms are necessary to reason about the domain being formalized.
- Convenience of the axiomatizer.
  - Modularity. The axiomatizer needs only to add new effect axioms.
  - Accuracy. No inadvertent omissions of frame axioms.
- For theorizing about actions.

Determinate Actions Only

- Recall the special syntactic form of effect axioms:

$$R(\vec{x}, s) \supset (\neg)F(\vec{x}, do(A, s)).$$

$$R(\vec{x}, y, s) \supset f(\vec{x}, do(A, s)) = y.$$

- These preclude *indeterminate* actions:

$$heads(do(flip, s)) \vee tails(do(flip, s)),$$

$$(\exists x) holding(x, do(pick\_up\_a\_block, s)).$$

- *Determinate action* = action whose effect axioms all have the syntactic form above for every fluent.
- Indeterminate actions: later.
- Determinate actions: Intuitively, complete information about the initial situation
  $\Longrightarrow$
  Complete information about the next situation.

Primitive Actions Only

- We as yet have no constructs for complex actions. All actions are *primitive*.
- Complex actions:

$$\textbf{if } car\_in\_driveway \textbf{ then } drive \textbf{ else } walk.$$

$$clear\_table \triangleq \textbf{while } [(\exists block)ontable(block)]\ remove\_a\_block.$$

$$remove\_a\_block \triangleq (\pi\ block)[pickup(block); put\_on\_floor(block)].$$

- Complex actions: later.

Limitations of This Version of the SC

- No time. So can't talk about how long actions take, when they occur.
- No concurrency.
- Discrete situation. No continuous actions like pushing an object from A to B.

A SOLUTION TO THE FRAME PROBLEM (SOMETIMES)

- "Sometimes" $=$ determinate actions $without$ ramifications (state constraints). Ramifications, later. Indeterminate actions, later.

Frame Axioms: Pednault's Proposal

- Assume given a set of positive and negative effect axioms (one for each action A and fluent F):
$$\begin{aligned} \varepsilon_F^+(\vec{x},\vec{y},s) &\supset F(\vec{x},do(A(\vec{y}),s)), \\ \varepsilon_F^-(\vec{x},\vec{y},s) &\supset \neg F(\vec{x},do(A(\vec{y}),s)). \end{aligned} \tag{1}$$
Here, $\varepsilon_F^+(\vec{x},\vec{y},s)$ and $\varepsilon_F^-(\vec{x},\vec{y},s)$ are first order formulas whose free variables are among $\vec{x},\vec{y},s$. Notice that in these effect axioms the variable sequences $\vec{x}$ and $\vec{y}$ consist of distinct variables.

- Completeness Assumption for Fluent Preconditions:
The fluent precondition $\varepsilon_F^+(\vec{x},\vec{y},s)$ specifies all the conditions under which action $A(\vec{y})$, if performed, will lead to the truth of F for $\vec{x}$ in A's successor situation. Similarly, $\varepsilon_F^-(\vec{x},\vec{y},s)$ specifies all the conditions under which action $A(\vec{y})$, if performed, will lead to the falsity of F for $\vec{x}$ in A's successor situation.

- By the completeness assumption, reason as follows:
Suppose that both $F(\vec{x},s)$ and $\neg F(\vec{x},do(A(\vec{y}),s))$ hold. Then F, which was true in situation s, was made false by action A. By the completeness assumption, the only way F could become false is if $\varepsilon_F^-(\vec{x},\vec{y},s)$ were true.

- This intuition can be expressed axiomatically by:
$$F(\vec{x},s) \wedge \neg F(\vec{x},do(A(\vec{y}),s)) \supset \varepsilon_F^-(\vec{x},\vec{y},s).$$
This is logically equivalent to:
$$F(\vec{x},s) \wedge \neg \varepsilon_F^-(\vec{x},\vec{y},s) \supset F(\vec{x},do(A(\vec{y}),s)).$$

- A symmetric argument yields the axiom:

$$\neg F(\vec{x}, s) \wedge \neg\varepsilon_F^+(\vec{x}, \vec{y}, s) \supset \neg F(\vec{x}, do(A(\vec{y}), s)).$$

- These have precisely the syntactic forms of positive and negative frame axioms and, by virtue of the argument leading to these, they play exactly the role of frame axioms.

- Provided the completeness assumption is true, there is a systematic way of obtaining the frame axioms from the effect axioms.

- **Example:**
  Recall,

$$fragile(x) \supset broken(x, do(drop(r, x), s)).$$

- Rewrite this into a logically equivalent form to conform to the pattern (1):

$$x = y \wedge fragile(x) \supset broken(x, do(drop(r, y), s)).$$

- The completeness assumption for this setting is that the only precondition for x being broken as a result of dropping y is that x be fragile and the same as y. This assumption yields the negative frame axiom for the fluent $broken$ wrt the action $drop$:

$$\neg broken(x, s) \wedge \neg[x = y \wedge fragile(x)] \\ \supset \neg broken(x, do(drop(r, y), s)).$$

- To obtain the frame axioms in this way for all fluent-action pairs requires considering a large number of "vacuous" effect axioms. For example, to obtain a frame axiom for the fluent $color$ wrt action $drop$, consider the positive effect axiom for $color$ wrt $drop$:

$$false \supset color(x, c, do(drop(r, y), s)).$$

From this we obtain the negative frame axiom for $color$ wrt $drop$:

$$\neg color(x, c, s) \supset \neg color(x, c, do(drop(r, y), s)).$$

- This illustrates two problems with Pednault's proposal:
  - To systematically determine the frame axioms for all fluent-action pairs from their effect axioms, we must enumerate (or at least consciously consider) all these effect axioms, including the "vacuous" ones. In particular, we must enumerate all fluent-action pairs for which the action has no effect on the fluent's truth value, which really amounts to enumerating most of the frame axioms directly.
  - The number of frame axioms so obtained is $2 \times \mathcal{A} \times \mathcal{F}$, where $\mathcal{A}$ is the number of actions, and $\mathcal{F}$ the number of fluents. Some of these may be vacuously true (i.e., when the fluent precondition of the corresponding effect axiom is $true$), but in general, we are faced with the usual difficulty associated with the frame problem – too many frame axioms.
- Summary: Pednault's proposal.

– It provides a systematic (and easily and efficiently implementable) mechanism for generating frame axioms from effect axioms.
– But it does not provide a parsimonious representation of the frame axioms.

Frame Axioms: Schubert's Proposal

- Schubert, elaborating on a proposal of Haas, argues in favor of what he calls *explanation closure axioms* for representing the usual frame axioms.
- Consider the fluent $holding$, and suppose that both $holding(r, x, s)$ and $\neg holding(r, x, do(a, s))$ are true. How can we explain the fact that $holding$ ceases to be true? If we assume that the only way this can happen is if the robot r put down or dropped x, we can express this with the explanation closure axiom:

$$holding(r, x, s) \wedge \neg holding(r, x, do(a, s)) \\ \supset a = putdown(r, x) \vee a = drop(r, x).$$

- Notice that this sentence quantifies universally over a (actions).
- To see how this functions as a frame axiom, rewrite it in the logically equivalent form:

$$holding(r, x, s) \wedge a \neq putdown(r, x) \wedge a \neq drop(r, x) \\ \supset holding(r, x, do(a, s)). \tag{2}$$

This says that all actions other than $putdown$ and $drop$ leave $holding$ invariant,[1] which is the standard form of a frame axiom (actually, a set of frame axioms, one for each action distinct from $putdown$ and $drop$).

[1] To accomplish this, we shall require unique names axioms like $pickup(r, x) \neq drop(r', x')$. We shall explicitly introduce these later.

- In general, an *explanation closure axiom* has one of the two forms:

$$F(\vec{x}, s) \wedge \neg F(\vec{x}, do(a, s)) \supset \alpha_F(\vec{x}, a, s),$$
$$\neg F(\vec{x}, s) \wedge F(\vec{x}, do(a, s)) \supset \beta_F(\vec{x}, a, s).$$

- In these, the action variable a is universally quantified. These say that if ever the fluent F changes truth value, then α_F or β_F provides an exhaustive explanation for that change.

- As before, to see how explanation closure axioms function like frame axioms, rewrite them in the logically equivalent form:

$$F(\vec{x}, s) \wedge \neg \alpha_F(\vec{x}, a, s) \supset F(\vec{x}, do(a, s)),$$

and

$$\neg F(\vec{x}, s) \wedge \neg \beta_F(\vec{x}, a, s) \supset \neg F(\vec{x}, do(a, s)).$$

- These have the same syntactic form as frame axioms with the important difference that action a is universally quantified. Whereas there would be $2 \times \mathcal{A} \times \mathcal{F}$ frame axioms, there are just $2 \times \mathcal{F}$ explanation closure axioms. This parsimonious representation is achieved by quantifying over actions in the explanation closure axioms.

- Schubert proposes that explanation closure axioms must be provided independently of the effect axioms. Like the effect axioms, these are domain-dependent. In particular, Schubert argues that they cannot be obtained from the effect axioms by any kind of systematic transformation. Thus, Schubert and Pednault entertain conflicting intuitions about the origins of frame axioms.

- Schubert's appeal to explanation closure as a substitute for frame axioms involves an assumption.

  The Explanation Closure Assumption

  The only way the fluent F's truth value could have changed from true to false under action a is if α_F were true. This means, in particular, that α_F completely characterizes all those actions a that can lead to this change; similarly for β_F.

- We can see clearly the need for this assumption from the example explanation closure axiom (2). If, in the intended application, there were an action (say, $eat(r, x)$) that could lead to r no longer holding x, axiom (2) would be false.

- **Summary: Schubert's Proposal**
  - Explanation closure axioms provide a compact representation of frame axioms – $2 \times \mathcal{F}$ of them. (Aside: This assumes they aren't too long. See later for an argument why they are likely to be short.)
  - But Schubert provides no systematic way of automatically generating them from the effect axioms. In fact, he argues this is impossible.

- Can we combine the best of Pednault and Schubert?

A Simple Solution to the FP (Sometimes)

- **Example**: Suppose there are two positive effect axioms for the fluent $broken$:

$$fragile(x) \supset broken(x, do(drop(r, x), s)),$$
$$nexto(b, x, s) \supset broken(x, do(explode(b), s)).$$

  These can be rewritten in the logically equivalent form:

$$\begin{array}{l} [(\exists r)\{a = drop(r, x) \wedge fragile(x)\} \\ \qquad \vee(\exists b)\{a = explode(b) \wedge nexto(b, x, s)\}] \\ \qquad\qquad \supset broken(x, do(a, s)). \end{array} \tag{3}$$

- Similarly, consider the negative effect axiom for $broken$:

$$\neg broken(x, do(repair(r, x), s)).$$

  In exactly the same way, this can be rewritten as:

$$(\exists r)a = repair(r, x) \supset \neg broken(x, do(a, s)). \tag{4}$$

- Now appeal to the following completeness assumption: *Axiom (3) characterizes all the conditions under which action a leads to y being broken.*

- Then if $\neg broken(x, s), broken(x, do(a, s))$ are both true, the truth value of $broken$ must have changed because

$$\begin{array}{l} (\exists r)\{a = drop(r, x) \wedge fragile(x)\} \\ \qquad \vee(\exists b)\{a = explode(b) \wedge nexto(b, x, s)\} \end{array}$$

  was true.